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Subject : EDS

The image displays two screenshots of the CODETANTRA online programming environment, showing solutions for two different problems.

Top Screenshot: 1.1.1. Calculate Momentum

The problem description states: "Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula: $p = m \times v$ ".

The input format specifies: "A single floating-point number representing the mass of the object in kilograms." and "A single floating-point number representing the velocity of the object in meters per second."

The output format specifies: "The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places)."

The sample test cases section is empty.

The solution code is as follows:

```
1 # Accept mass and velocity as input
2 mass = float(input())
3 velocity = float(input())
4
5 # calculate momentum
6 momentum = mass * velocity
7
8 # display result rounded to 2 decimal places with unit
9 print(f"momentum:.2f)kgm/s")
```

The execution results show: "2 out of 2 shown test case(s) passed" and "2 out of 2 hidden test case(s) passed".

Bottom Screenshot: 1.1.2. Conditional Calculation Based on the Number of Digits

The problem description states: "Write a Python program that accepts an integer n as input. Depending on the number of digits in n ."

The constraints are: "1 ≤ n ≤ 999".

The input format specifies: "The input consists of a single integer n ."

The output format specifies: "If n is a single-digit number, print its square." "If n is a two-digit number, print its square root (rounded to two decimal places)." "If n is a three-digit number, print its cube root (rounded to two decimal places)." "Else print 'Invalid'."

The sample test cases section is empty.

The solution code is as follows:

```
1 import math
2
3 # Accept input
4 num = int(input())
5
6 # Check number of digitals and perform StopIteration
7 if 1 <= num <= 9:
8     print(num ** 2)
9
10 elif 10 <= num <= 99:
11     print(f"{math.sqrt(num):.2f}")
12
13 elif 100 <= num <= 999:
14     print(f"{math.cbrt(num):.2f}")
15
16 else:
17     print("Invalid")
```

The execution results show: "4 out of 4 shown test case(s) passed" and "3 out of 3 hidden test case(s) passed".

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 number = int(input())
2 reversed = 0
3
4 while number > 0:
5     digit = number % 10
6     reversed = reversed * 10 + digit
7     number = number // 10
8
9 print(reversed)
10
```

Average time0.001 s1.40 ms

Maximum time0.002 s2.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output

5387

7635

Actual output

5387

7635

Test case 2

Terminal

Test cases

Prev

Reset

Submit

Next

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1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

The first line contains the first date in the format *YYYY-MM-DD*.
The second line contains the second date in the format *YYYY-MM-DD*.

Output Format:

An integer representing the number of days between the given dates.

Note:

The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import date
2
3 from datetime import datetime
4
5 date1 = input().strip()
6 date2 = input().strip()
7
8 d1 = datetime.strptime(date1, "%Y-%m-%d")
9 d2 = datetime.strptime(date2, "%Y-%m-%d")
10
11 difference = d2 - d1
12
13 print(difference.days)
14
```

Average time0.007 s7.00 ms

Maximum time0.020 s20.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2

Terminal

Test cases

Prev

Reset

Submit

Next

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1.1.8. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

The character "x" is used in the output to represent multiplication.
Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

```
1 num = int(input())
2 for i in range(1,11):
3     print(f"{num} x {i} = {num * i}")
4
```

Average time0.003 s2.50 msMaximum time0.004 s4.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1

Expected output

8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40

Actual output

8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40

TerminalTest cases

PrevResetSubmitNext

Lab Assignment :

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses. The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorfa...

```
16 else:
17     # Calculate aggregate percentage
18     aggregate = sum(marks) / n
19
20     # Print aggregate percentage (2 decimal places)
21     print(f"Aggregate Percentage: {aggregate:.2f}")
22
23     # Determine grade
24     if aggregate > 75:
25         print("Grade: Distinction")
26     elif 60 <= aggregate < 75:
27         print("Grade: First Division")
28     elif 50 <= aggregate < 60:
29         print("Grade: Second Division")
30     elif 40 <= aggregate < 50:
31         print("Grade: Third Division")
```

Average time0.003 sMaximum time0.012 s3.10 ms12.00 ms5 out of 5 shown test case(s) passed5 out of 5 hidden test case(s) passed

Test case 1412 ms

Expected output455 65 78 89Aggregate Percentage: 71.75Grade: First DivisionActual output455 65 78 89Aggregate Percentage: 71.75Grade: First Division

Test cases

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     elif n == 2:
5         return 1
6     else:
7         # write the code..
8
9     return fibonacci(n-1) + fibonacci(n-2)
10
11
12 n = int(input())
13 for i in range(1, n + 1):
14     print(fibonacci(i), end=" ")
15
```

Average time0.006 sMaximum time0.019 s6.50 ms19.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1412 ms

Expected output50 1 1 2 3Actual output50 1 1 2 3

Test case 2412 ms

Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 # Input: number of rows
2 n = int(input())
3
4 # Print pattern
5 for i in range(1, n + 1):
6     for j in range(i):
7         print("*", end=" ")
8     print()
```

Average time0.002 sMaximum time0.003 s1.83 ms3.00 ms2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 1412 ms

Expected output5* * * * *Actual output5* * * * *

Test cases

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Input: number of rows
2 n = int(input())
3
4 # Print pattern
5 for i in range(1, n + 1):
6     for j in range(1, i + 1):
7         print(j, end=" ")
8     print()
```

Average time0.002 s1.50 msMaximum time0.002 s2.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 1

Expected output

1-
1 2-
1 2 3-
1 2 3 4-
1 2 3 4 5-

Actual output

1-
1 2-
1 2 3-
1 2 3 4-
1 2 3 4 5-

TerminalTest cases

< PrevResetSolveNext >